

Problem

The loop equation for a series RLC circuit gives

$$\frac{di}{dt} + 2i + \int_{-\infty}^t i dt = \cos 2t \text{ A}$$

Assuming that the value of the integral at $t = -\infty$ is zero, find $i(t)$ using the phasor method.

Step-by-step solution

Step 1 of 2

$$\frac{di}{dt} + 2i + \int_{-\infty}^t i dt = \cos 2t$$

Let $i = I \cos(\omega t + \phi)$

$$\frac{di}{dt} + 2i + \int_{-\infty}^t i dt = \cos 2t .$$

Assuming integral at $t = -\infty$ is zero.

Step 2 of 2

Representing in frequency domain,

$$j\omega I + 2I + \frac{I}{j\omega} = 1\angle 0^\circ$$

$$\because \omega = 2$$

$$j2I + 2I - j\frac{I}{2} = 1\angle 0^\circ$$

$$I\left[2 + 2j - \frac{j}{2}\right] = 1\angle 0^\circ$$

$$I[2 + 1.5j] = 1\angle 0^\circ$$

$$I = \frac{1\angle 0^\circ}{2.5\angle 36.869^\circ}$$

$$I = 0.4\angle -36.869^\circ .$$

In time domain,

$$i(t) = 0.4\cos(2t - 36.869^\circ) .$$